RESUME

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EDUCATION

Nanyang Technological University, Singapore — Aug 2022 – Jun 2026
Bachelor of Computing in Computer Science, College of Computing and Data Science
Honours(Merit) with a specialization in Data Science

PUBLICATIONS

Name in bold denotes the author.

- **Jiahua Zhang**, et al. *Process Reward Model as Cross-Task Learner: An Extension of Vision-Language Model to Zero-Shot Medical Reasoning*. Proceedings of the 34th ACM International Conference on Multimedia (**ACM MM**), 2026.
- **Jiahua Zhang**, J. Liang, Y. Cai, J. Ni, J. He. *Can LLM Understand Medical Imaging with Long Context?* Proceedings of the IEEE International Conference on Multimedia and Expo (**ICME**), 2026. **Oral**.
- **Jiahua Zhang**, Y. Tian. *DiffiT-HSFDA: Diffusion Based Source-Free Domain Adaptation for Histopathology*. Advanced Intelligent Computing Technology and Applications (**ICIC**), Springer, 2025. **Oral**. DOI
- C. Zhang, X. Wang, H. Zhang, **Jiahua Zhang**, et al. *LayerLog: Log Sequence Anomaly Detection Based on Hierarchical Semantics*. Applied Soft Computing (**ASOC**), 2023. Impact Factor 7.8, **JCR Q1** . DOI
- C. Zhang, X. Wang, **Jiahua Zhang**, et al. *VESC: A New Variational Autoencoder Based Model for Anomaly Detection*. International Journal of Machine Learning and Cybernetics (**IJMLC**), 2022. Impact Factor 2.9. DOI

RESEARCH EXPERIENCE

Guangzhou Medical University — State Key Laboratory of Respiratory Disease — Feb 2024 – Present
Research Collaborator, Thoracic Surgery Department and Thoracic Oncology Group

- Collaborated with Professor He Jianxing's thoracic surgery team on a visual foundation model programme for lung-cancer diagnosis from histopathology and computed tomography.
- Investigated region-of-interest activation failure in weakly supervised segmentation of lung cancer histopathology under cross-domain transfer with varying staining protocols, and designed a diffusion based source free domain adaptation pipeline that restored high accuracy segmentation without source data or additional annotation.
- Developed training-free multimodal large language model frameworks for computed tomography and whole-slide image analysis, adapting long-context models to high-dimensional medical imaging without task-specific fine-tuning.
- **Achievements:** one oral paper at ICIC 2025 and one oral paper at IEEE ICME 2026.

Nanyang Technological University — CAO Exchange Programme, Shenzhen — Nov 2025 – Feb 2026
Research Intern — BioMedPRM: Process Reward Modelling for Biomedical Reasoning

- Investigated step-level reward modelling to improve the reliability of large language model reasoning chains in biomedical question answering, supervising each reasoning step rather than only the final answer.
- Designed and evaluated a process reward model that transfers across tasks, with emphasis on robustness under distribution shift and in out-of-distribution clinical scenarios.
- **Achievements:** one paper accepted at ACM Multimedia 2026.

Nanyang Technological University — School of Social Sciences (Psychology) — Oct 2024 – May 2025
Research Assistant, supervised by Assistant Professor Luo Lizhu

- Represented high dimensional biomedical data, including functional MRI and whole slide images, as graphs so that language models can reason over inputs carrying far more variables than a context window admits.
- Developed a training-free graph in-context learning method for medical multimodal prediction and benchmarked it against fine-tuned baselines.

Pohang University of Science and Technology (POSTECH) — DPNM Laboratory — Jun 2024 – Aug 2024
POSTECH Summer Program (PSP), supervised by Professor James Won-Ki Hong

- Designed a convolutional diffusion-denoising pipeline for MRI and X-ray segmentation and spinal disease classification, improving performance on the BUU and lumbar-spine MRI datasets.
- Applied representation learning and natural language processing to hospital chief-complaint records, evaluating clustering and vectorisation methods including TF-IDF, and developed lightweight algorithms for edge-computing.

Nanyang Technological University — S-Lab for Advanced Intelligence — Mar 2024 – Jun 2024
Student Research Assistant, LLM-based video generation

- Improved the visual quality of AI-generated video by developing and applying noise-detection algorithms.
- Built label preprocessing for the performance evaluation benchmark used to assess generated video quality.

Harbin Institute of Technology, Shenzhen — Jan 2022 – Jul 2022
Research Assistant

- Carried out the data-processing component of two anomaly-detection studies, covering cleaning, preprocessing and preparation of large-scale log and time-series datasets for model training and evaluation.
- Produced the results visualisation for both studies, including the comparative and ablation figures reported in the published analyses.
- **Achievements:** co-authored two journal papers.

ENGINEERING AND INDUSTRY EXPERIENCE

Open-Source Speech Synthesis Toolkit and Hosted Inference Service 2022 – 2023
Independent developer

- Trained speech-synthesis models and released a repository integrating VITS speech-synthesis APIs with large language models, adopted by content creators for video production and livestreaming. Repository
- Deployed and maintained a public inference platform on Hugging Face containers, self-funding GPU serving and optimising latency, cost and concurrent load. Demo

Clangorous Technological Company — Intern, Full-Stack Developer Dec 2023 – Feb 2024

- Contributed to development and testing of .NET, MVC and ERP modules across the full stack.
- Built and released a full-stack application project. Repository

TECHNICAL SKILLS AND COMPETENCIES

- **Machine learning:** PyTorch, Hugging Face Transformers, PyTorch Geometric, torchvision, torchaudio; large-model training and serving on multi-GPU cluster infrastructure.
- **Medical and multimodal AI:** medical image analysis (CT, MRI, X-ray, whole-slide imaging, functional MRI), vision-language models, process reward modelling, domain adaptation and transfer learning, test-time computation.
- **Software and web development:** Python, C# and .NET, ABP, Flask, Jinja2, Vue, React, MySQL, Android, Unity.
- **Data engineering:** data acquisition and web scraping, preprocessing pipelines for large-scale clinical and imaging data.
- **Languages:** Mandarin Chinese (native), English (IELTS 7.0).

VOLUNTEER AND COMMUNITY ENGAGEMENT

NTU Chinese Society — Committee Member, Spring Festival Gala 2025

- Volunteered the 2025 NTU Spring Festival Gala, the society's largest annual stage production, presenting the full programme to a student and community audience.

NTU Chinese Society — Committee Member, Publicity Team 2023 – 2024

- Served as event manager within the publicity team, writing and typesetting articles for the society's WeChat official account to promote society events throughout the academic year.

NTU IET Club (Institution of Engineering and Technology) — Event Manager Aug 2022 – May 2023

- Led the organisation of IT-oriented events and technical competitions for club members and the wider NTU student body, coordinating a committee of 5 members across 3 events and taking responsibility for planning, publicity and delivery.

Shenzhen International High School Students Forum — Student Volunteer 2019

- Supported visiting delegates from different school systems in preparing and presenting their work, and assisted with on-site coordination of the forum.